

CXXXII.—*The Atomic Weight of Lead from Ceylon
Thorite.*

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THE question discussed by one of us (T., 1911, 99, 82) whether
some of the common elements may not in reality be mixtures of
chemically non-separable elements of different atomic weights, has
been advanced by the recent generalisation connecting the position

of the product of a radioactive change in the periodic table with that of the parent and with the nature of the ray expelled in the change. This showed that all the radio-elements which are chemically identical in character occupy the same place in the periodic table. Hence such elements are conveniently referred to as isotopic. The unknown end-products of all the known disintegration series fall, by the application of this generalisation, into the same place in the periodic table, namely, that occupied by lead. If they are entirely stable, that is, if the apparent ends of the disintegration series are the real ends, the lead from radioactive minerals should differ in atomic weight according to the proportion of uranium and thorium in the mineral. The calculated atomic weight of the isotope derived from uranium is 206, and that of the isotope derived from thorium is 208.4, since they are formed from radium, 226.0, and thorium, 232.4, by the loss of five and six atoms of helium respectively (K. Fajans, *Physikal. Zeitsch.*, 1913, **14**, 140; F. Soddy, *Chem. News*, 1913, **107**, 97).

Boltwood, to whom the original suggestion that lead may be the end-product of uranium is due (*Amer. J. Sci.*, 1907, [iv], **23**, 77), found the ratio of lead to uranium in primary minerals from the same locality to be nearly constant. Both he and, later, Holmes (*Proc. Roy. Soc.*, 1911, *A*, **85**, 248) considered it highly improbable that thorium should also give lead. The question is, however, still an open one. It is complicated by the possibility that thorium may, in some yet unknown way, be genetically connected with the uranium series, and by the slow rate of change of thorium in comparison with that of uranium. A recent estimate of the ratio of the rates of change is 0.36 (McCoy, *Physical Review*, 1913, 393), and this makes the uranium in a mineral about three times as effective in producing lead as the thorium.

An attempt has been made to obtain direct experimental evidence on the point at issue. For this purpose Ceylon thorite, a hydrated silicate of thorium discovered at the same time as the more common thorianite, is uniquely suited. According to the original analysis (W. R. Dunstan, *Ceylon Mineralogical Survey Report*, March 31st, 1904; *Nature*, 1904, **69**, 510), lead is recorded as absent, whilst the thorium and uranium contents are respectively $\text{ThO}_2 = 66.26$ and $\text{UO}_3 = 0.46$ per cent. We have found, however, quite a detectable quantity of lead in the mineral, and from the amount of thorite in our possession enough lead has been separated to enable comparative estimations of its atomic weight in terms of that of ordinary lead to be made.

Analysis of the Mineral.

The mineral is readily decomposed by hydrochloric acid. The undissolved part was treated with hydrofluoric acid, and the silica estimated by difference, the small residue remaining not being further examined. The lead was separated from the dilute hydrochloric acid solution by hydrogen sulphide. The precipitate, obtained from the filtrate by ammonia, was redissolved in acid, and reprecipitated by oxalic acid. The precipitate was treated with ammonium oxalate to dissolve thorium and zirconium, and from the solution the thorium was precipitated by hydrogen peroxide. The filtrate from the oxalic acid precipitate was poured into excess of ammonium carbonate, and the uranium estimated in the filtrate.

Separate estimations were also made of the thorium, uranium, and lead. A determination of the helium in the mineral is not yet completed. The analysis is shown below:

ThO ₂ .	SiO ₂ .	Fe ₂ O ₃ .	Ce ₂ O ₃ .	ZrO ₂ .	P ₂ O ₅ .	Al ₂ O ₃ .	U ₃ O ₈ .	CaO.	PbO.
61.95	11.91	5.18	4.99	3.92	0.66	0.80	0.85	0.40	0.39
		H ₂ O.	Insoluble residue.			Total.			
		7.06	1.80			99.91 per cent.			

With regard to the uranium, there are difficulties in its chemical estimation in the presence of so large a preponderance of thorium. A preliminary test of the amount of radium in the mineral, by the emanation method, indicates a larger proportion of uranium than is shown in the chemical analysis, namely, 1.9 per cent. of U₃O₈. This is the value provisionally adopted, rather than that of the chemical analysis.

It will be seen from the proportions of the significant constituents, thorium, uranium, and lead, that the mineral is an unusually favourable one for the examination. The ratio of uranium to thorium is certainly not greater than 0.03, so that if the lead in the mineral is entirely of radioactive origin, about ten parts of it will be derived from the thorium for one part from the uranium. Its atomic weight should therefore be about 208.2. On the other hand, although the small quantity of lead in the mineral makes the work laborious, it is favourable to the possibility that it may all be of radioactive origin, and that none of it was an original constituent. More definite evidence on this point will be available, however, when the helium and radium estimations are concluded.

Extraction and Purification of the Lead.

The lead sulphide precipitate from about a kilogram of the mineral, after washing and digestion with ammonium sulphide, was

dissolved in nitric acid and precipitated as sulphate. The ammonium acetate solution of the sulphate was reprecipitated by hydrogen sulphide, and the precipitate again dissolved in nitric acid. The lead was precipitated as iodide, washed, and crystallised from about two litres of hot water. From the ammonium acetate solution of the iodide the lead was separated as sulphide, converted into nitrate, finally precipitated as chloride, washed, and crystallised from hot water, and dried at 100° . The mother liquor was kept distinct. About 1.2 grams of the purified lead chloride was so obtained.

Ordinary lead nitrate, after three recrystallisations, was converted into iodide, and thenceforth treated exactly in the same manner as the thorite lead, being finally obtained as chloride.

The Relative Atomic-weight Estimation.

The atomic-weight estimations were purely relative, exactly similar determinations being made with the two preparations of lead chloride described, and the atomic weight of the thorite lead was calculated from that of ordinary lead, taken as 207.1. The method of weighing the lead chloride was similar to that adopted by Baxter and Wilson in their atomic-weight determinations, on which, mainly, the international value is based. The dry lead chloride was fused in a platinum boat in a current of dried hydrogen chloride, in which it was allowed to cool before displacing the hydrogen chloride with a current of dried air. The empty boat before and after the experiment was treated in an identical manner. After weighing, the boat and the fused lead chloride were placed in a stoppered bottle with a litre of water containing 5 c.c. of dilute nitric acid, and gently rocked on a shaking machine until the lead chloride was dissolved to a completely clear solution. The boat was carefully washed, dried at 100° , and heated in hydrogen chloride as at first. In no case did its final weight differ by more than 0.2 mg. from its initial weight.

The lead chloride solution was titrated with silver nitrate solution in the dark room by the aid of a bright red light placed alongside. Two hundred c.c. of the silver solution were run in by a pipette, followed by a quantity from a burette (about 10 c.c.), estimated to be less than the required amount by 1 c.c. The contents of the bottle were well shaken and allowed to settle, a drop of the silver solution added from the burette, and the time of appearance of the cloud observed. At first the cloud would be visible after a few seconds from the addition of the drop, but as the end-point was approached, the time of appearance rapidly

lengthened to several minutes. The uncertainty in the end-point was not greater than a single drop, or 1 part in 4000.

When 1 drop no longer produced a visible cloud after five minutes, 5 drops added together would produce a cloud after about a minute. A second addition of 5 drops then produced no further cloud. This uncertainty, although it would affect an absolute determination, can have no effect on a purely relative one. The time of the appearance of the drop is also a valuable independent check on the approach of the end-point. The solubility of the silver chloride, again, which would be of very great importance in any absolute determination, is without appreciable effect in the relative determination.

The same platinum boat, the same weights, the same pipette, burette, and silver nitrate solution were used throughout each estimation, which consisted of three titrations, one with ordinary lead chloride before and one after the thorite lead titration.

After the first estimation the lead was recovered from all three solutions, and used for the second set. The solutions, containing an excess of chlorine, were filtered, and evaporated to dryness, the residue being redissolved in dilute nitric acid, and the lead precipitated from the solution as sulphate. This was dissolved in ammonium acetate solution, precipitated as sulphide, which, after washing and decomposition with hydrochloric acid, was filtered, evaporated to small bulk, added to the final mother liquor obtained in the previous experiment, and allowed to crystallise as before.

Results.

The results so far obtained are tabulated below. The first column gives the weight and kind of lead chloride used, the second the volume of the silver solution required, and the third the volume of solution per gram of lead chloride. The fourth column contains the ratio of the volume for the ordinary lead chloride in comparison with that for the thorite lead chloride as unity. The last column gives the relative atomic weight of thorite lead in terms of that of ordinary lead. The results, all except that marked with an asterisk, agree very closely, and, indeed, the closeness of the agreement is no doubt in some part accidental. The determination marked with an asterisk was the last to be performed, and was not done on the same day as the other two in this experiment. It differs from the first result by 1 part in 800.

Table of Results.

	Weight of PbCl ₂ (grams).	Volume of AgNO ₃ soln. used (c.c.).	1 Gram of PbCl ₂ requires	Ratio.	At. wt. of thorite lead.
Experiment I.	Ordinary :				
	(a) 1.0950	209.86	191.65		
	(c) 1.0912	209.10	191.62		
			Mean 191.64	1.0049	208.5
	Thorite :				
	(b) 1.1127	212.19	190.70		
Experiment II.	Ordinary :				
	(a) 1.0957	210.90	192.48		
	* (c) 1.0986	211.20	192.24		
			Mean 192.36	1.0042	208.3
	Thorite :				
	(b) 1.0980	210.34	191.57		
			Mean 1.0046	208.4	

It will be seen that there is a difference in the mean result of 1 part in 225 in the volume of the silver solution required for equal weights of the two chlorides. This is certainly several times greater than the probable error of the experiment, and greater than the possible maximum error of a single experiment, unless unknown sources of error exist.

Although not yet sufficient to establish finally the point at issue, these preliminary results show clearly a difference in the atomic weights in the expected direction of the right order of magnitude. As some time must necessarily elapse before the research can be brought to a conclusion and before larger quantities of the mineral can be obtained, it seems advisable to publish what has already been found.

It may be noted in passing that out of some twenty-four determinations of the atomic weight of lead, referred to by Brauner (Abegg's "Handbuch," III., ii, 613), seven give values between 206.80 and 206.95, six between 206.95 and 207.10, none between 207.10 and 207.35, and six between 207.35 and 207.60, thus showing very large differences.

Spectrographic Examination.

A detailed spectrographic examination of the two specimens of lead used is being undertaken. The silver lines 2447.8, 2437.8, 2413.2 show faintly in the spark spectrum of both specimens, but about twice as strongly in the thorite as in the ordinary lead. The copper line 3247.6 is visible in the ordinary, and only very faintly

in the thorite lead. The impurities, however, so far as they have been identified, do not seem to be of great significance.

With regard to the lead lines, no differences of wave-length, within the power of the Féry spectrograph to disclose, have been observed for the two specimens. The sole difference, and it seems at first sight a rather remarkable one, is in the line 4760·1, which is much stronger in all the photographs of ordinary lead than in those of thorite lead, in which it frequently is invisible.

We desire to express our thanks to the Carnegie Trust for the Féry spectrograph, which was purchased out of a Research Grant.

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